



M V S R ENGINEERING COLLEGE

(An Autonomous Institution)

(Sponsored by Matrusri Education Society, Affiliated to Osmania University)

Nadergul, Hyderabad - 501510 website: www.mvsrec.edu.in

NBA Accredited B.E. Courses:- Civil, CSE, ECE, EEE, IT & Mechanical



Department of Computer Science and Engineering (Allied Branches)

PROJECT TITLE : “Smart waste Detector”

BE CSE-AI & ML –Sem IV 2025-2026:

Batch – No : 15

Student Team Members :

- 1) Gangireddy Sankeerthana , 2451-24-748-018
- 2) G Kavya , 2451-24-748-017
- 3) Merugumalla Deekshitha , 2451-24-748-042

Guide : B . Venkataramana, Assoc Prof, Dept . of CSE-Allied Branches ,MVSREC

Problem Statement :

Improper waste segregation and disposal contribute to environmental pollution, reduced recycling efficiency, and increasing landfill waste. Many people lack awareness about identifying waste types, proper disposal methods, and nearby recycling facilities. Municipal authorities also face challenges in locating and responding quickly to waste accumulation in public areas. Existing solutions provide limited citizen participation and sustainability guidance. Smart Waste Analyzer addresses these issues through AI-powered waste detection, recycling recommendations, community waste reporting, and eco-friendly sustainability solutions.

Project Scope :

The Smart Waste Analyzer uses AI to detect and classify waste from images and provide disposal, recycling, and sustainability recommendations. It helps users find nearby recycling centers, generate DIY reuse ideas, and report waste locations while earning eco-points. The platform also enables municipality workers to monitor and manage reported waste through an admin dashboard.

Project Objectives

- Develop an AI-powered waste classification system to accurately identify and categorize waste materials from images for proper segregation and disposal.
- Promote recycling and sustainable waste management by providing disposal guidelines, recycling recommendations, and eco-friendly reuse ideas.
- Enhance community and municipal participation through waste reporting, recycling center discovery, and an admin dashboard for efficient waste monitoring and management.

Application Areas	Challenges
Smart Cities – Enables efficient waste monitoring and management in urban areas.	Waste Classification Accuracy – Similar-looking waste items can cause misclassification.
Residential Communities – Assists households in proper waste segregation and disposal.	Dataset Availability – Large and diverse datasets are required for effective model training.
Educational Institutions – Promotes environmental awareness and sustainable practices.	Image Quality Issues – Poor lighting and blurred images can affect detection performance.
Commercial Buildings – Supports effective waste management in offices and businesses.	User Participation – Encouraging regular user engagement can be challenging.
Recycling Centers – Improves waste sorting and recycling operations.	Real-Time Monitoring – Managing and processing large numbers of waste reports efficiently is difficult.
Public Places – Helps identify and report waste accumulation in community spaces.	Data Maintenance – Keeping recycling center and disposal information updated requires continuous effort.

Software/Hardware requirements for development/deployment

Tools /Technology	Usage
1. React JS & Tailwind CSS, Axios	Builds responsive and modern user interface design. It sends API request to the backend.
2. Node.js & Express.js	Handles backend logic and API routing.
3. MongoDB & Mongoose	Manages database storage and structured data modeling.
4. APIs (Llama 3 (Groq API)& Google Maps API)	Enables intelligent sustainability recommendations and location-based recommendations
5. Ultralytics YOLO v8	Detects waste objects in images.
6. Custom Waste Dataset, Polinations AI	Provides domain-specific data for analysis and recommendations. DIY image generation.
7. OpenCV & TensorFlow	Processes images and performs waste classification.
8. Deployment	Versel, Render ,MongoDB Atlas.

Certifications

Certification-01 : Web Development Bootcamp



Certification-02: JavaScript Essentials 2



I/O Specifications :

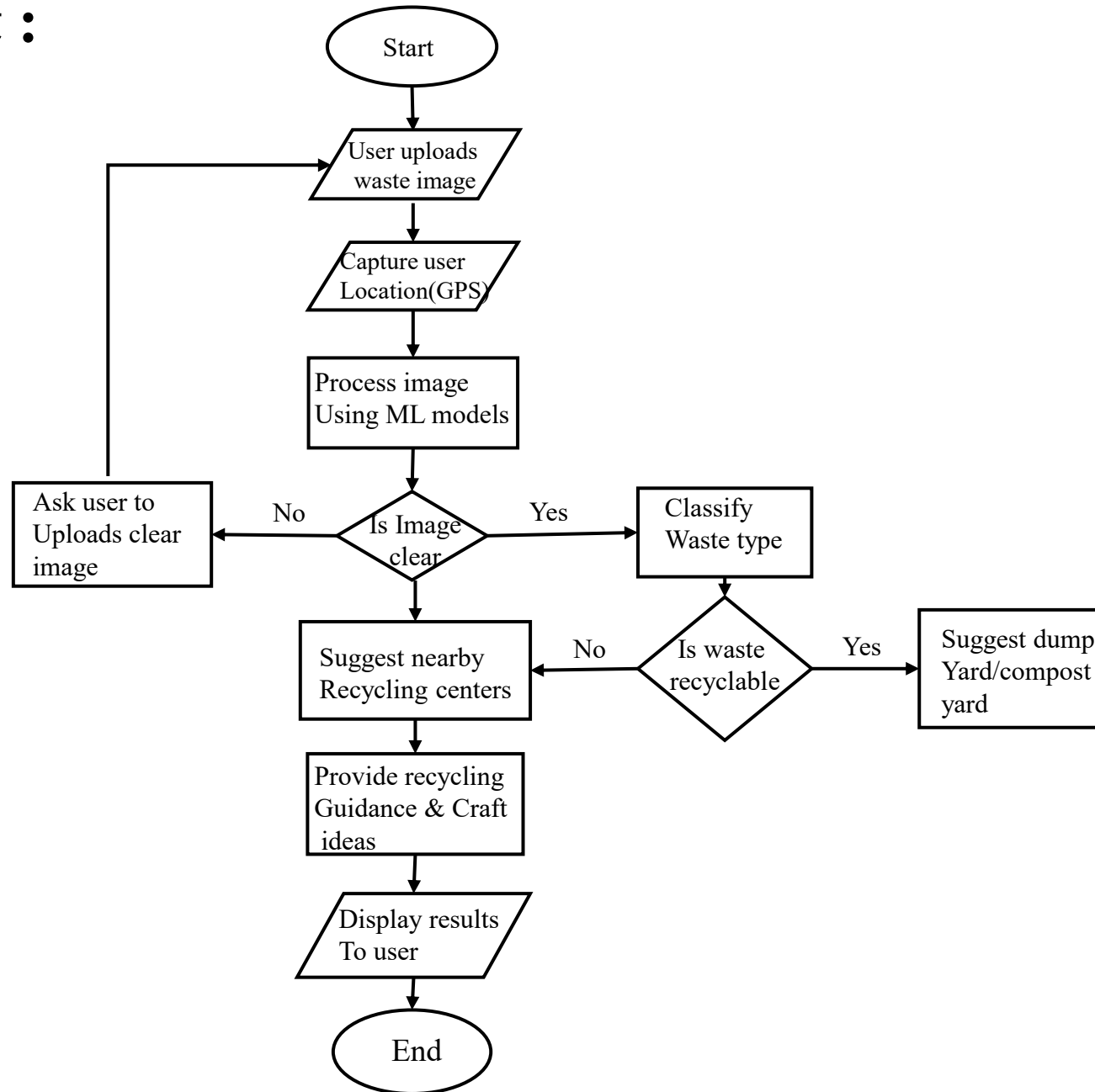
Input Specifications :

- Waste Image: Uploaded image or camera capture for analysis.
- User Location: GPS coordinates for recycling center recommendations.
- User Data: Login details for authentication and eco-point tracking.

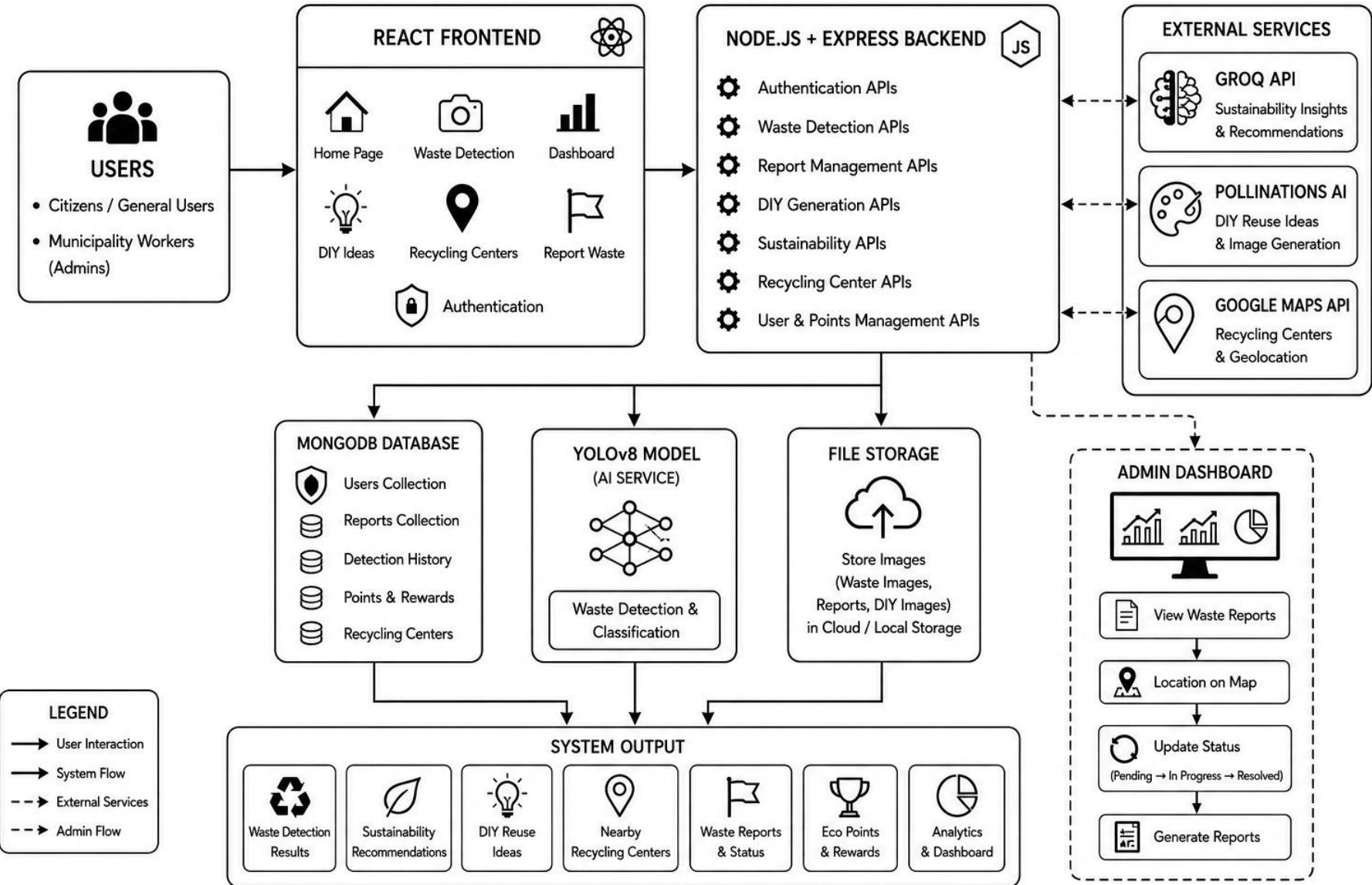
Output Specifications :

- Waste Detection: Identified waste types and object counts.
- Sustainability Insights: AI-generated disposal and recycling suggestions.
- DIY Ideas: Reuse recommendations with generated images.
- Recycling Centers: Nearby recycling locations with map navigation.
- Eco Reports: Waste reporting status and earned eco-points.

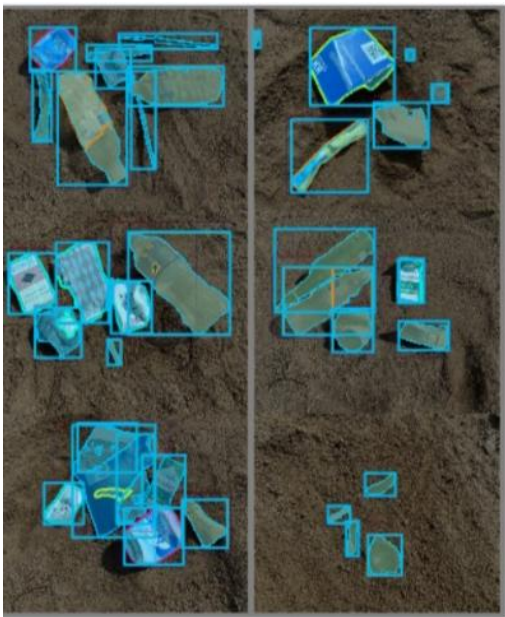
Flow Chart :



Proposed System Architecture :



AI Model (YOLOv8 + Roboflow) :



Metric	Value
Precision	84.28%
Recall	79.22%
mAP@0.5	86.26%
mAP@0.5	0.9572.14%

COLOR ?

CLASS NAME

COUNT ↻

cardboard583

glass734

metal761

paper402

plastic1,681

trash1,512

2445 Total Images

View All Images →

Dataset Split

TRAIN SET

70%

1722 Images

VALID SET

20%

481 Images

TEST SET

10%

242 Images

Preprocessing

Resize: Stretch to 640x640

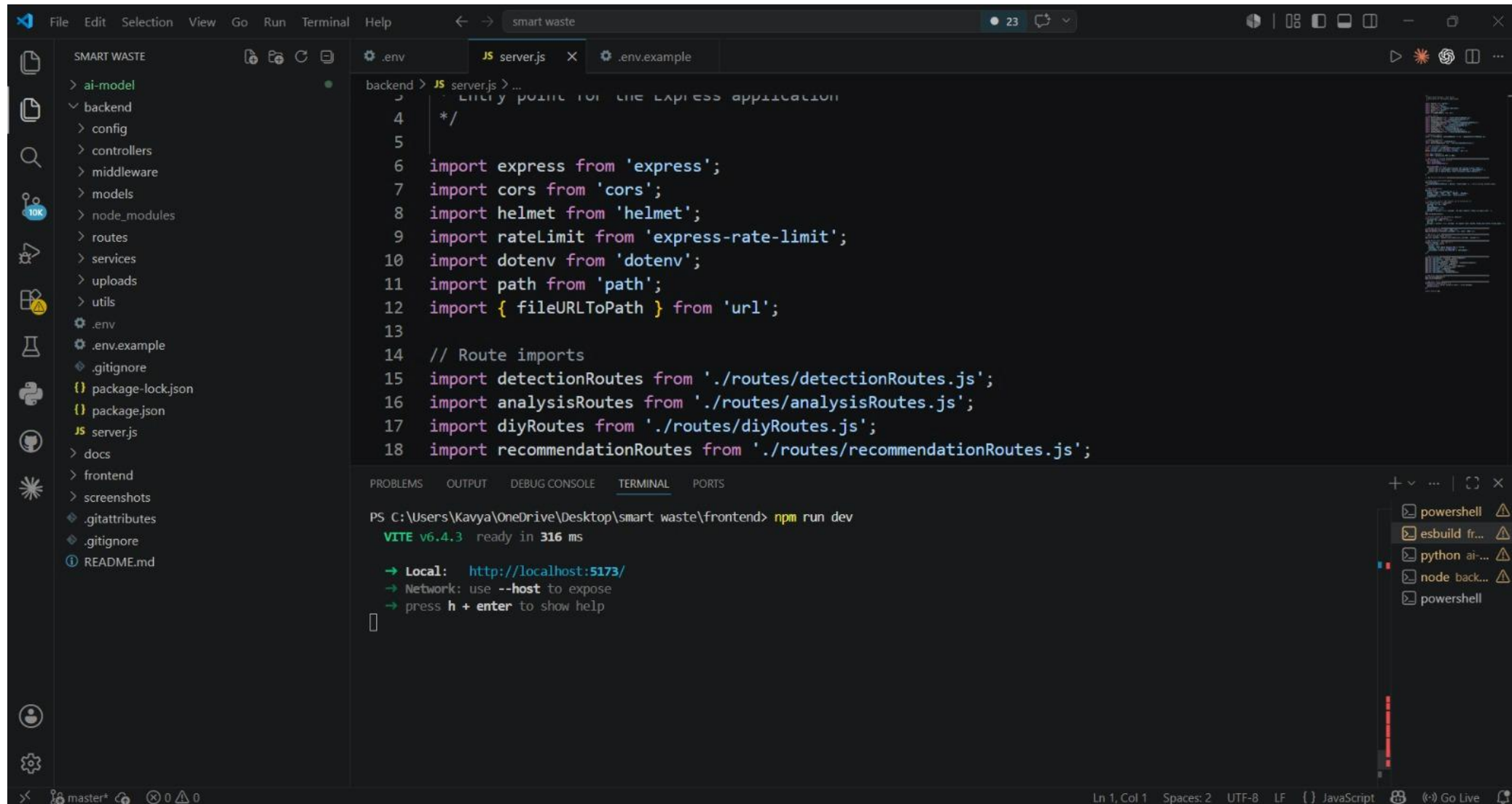
Augmentations

No augmentations were applied

Module Description :

Module	Description
1. User Authentication Module	Manages user registration, login, and secure access control for users and administrators.
2. Waste Detection Module	Uses YOLOv8 to detect and classify different types of waste from uploaded images or camera input.
3. Waste Analysis Module	Generates waste statistics, object counts, and category-wise distribution based on detection results.
4. Sustainability Insights Module	Provides AI-powered disposal guidance and 4R recommendations (Reduce, Reuse, Recycle, Recover).
5. Recycling Center Module	Locates nearby recycling centers and provides navigation through Google Maps integration.
6. Community Waste Reporting Module	Allows users to report waste locations, earn eco-points, and enables municipality workers to track and manage reports.

Implementation :



The screenshot shows the Visual Studio Code editor with a project named "smart waste". The file explorer on the left shows the project structure, including a "backend" directory. The "server.js" file is open in the editor, showing the following code:

```
backend > JS server.js > ...
1  // Entry point for the Express application
2
3  /*
4   */
5
6  import express from 'express';
7  import cors from 'cors';
8  import helmet from 'helmet';
9  import ratelimit from 'express-rate-limit';
10 import dotenv from 'dotenv';
11 import path from 'path';
12 import { fileURLToPath } from 'url';
13
14 // Route imports
15 import detectionRoutes from './routes/detectionRoutes.js';
16 import analysisRoutes from './routes/analysisRoutes.js';
17 import diyRoutes from './routes/diyRoutes.js';
18 import recommendationRoutes from './routes/recommendationRoutes.js';
```

The terminal at the bottom shows the command `npm run dev` being executed, resulting in the application running on `http://localhost:5173/`. The terminal output includes the message "VITE v6.4.3 ready in 316 ms" and instructions to press `h + enter` to show help.

```
=====
| No running processes found |
+-----+
```

▶ `from ultralytics import YOLO`

```
model = YOLO("yolov8n.pt")
```

```
results = model.train(
    data="/content/dataset/data.yaml",
    epochs=100,
    imgsz=640,
    batch=16,
    cache=True,
    patience=20
)
```

... Creating new Ultralytics Settings v0.0.6 file ✓

View Ultralytics Settings with 'yolo settings' or at '/root/.config/Ultralytics/settings.json'

Update Settings with 'yolo settings key=value', i.e. 'yolo settings runs_dir=path/to/dir'. For help see <https://docs.ultralytics.com/quickstart/#ultralytics-settings>.

Downloading <https://github.com/ultralytics/assets/releases/download/v8.4.0/yolov8n.pt> to 'yolov8n.pt': 100% — 6.2MB 362.4MB/s 0.0s

Ultralytics 8.4.66 🚀 Python-3.12.13 torch-2.11.0+cu128 CUDA:0 (Tesla T4, 14913MiB)

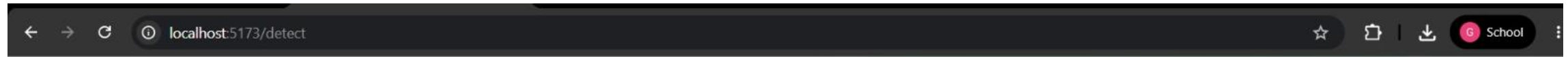
engine/trainer: agnostic_nms=False, amp=True, angle=1.0, augment=False, auto_augment=randaugument, batch=16, bgr=0.0, box=7.5, cache=True, cfg=None, classes=None, close_mosa:

Downloading <https://ultralytics.com/assets/Arial.ttf> to '/root/.config/Ultralytics/Arial.ttf': 100% — 755.1KB 123.3MB/s 0.0s

Overriding model.yaml nc=80 with nc=6

	from	n	params	module	arguments
0	-1	1	464	ultralytics.nn.modules.conv.Conv	[3, 16, 3, 2]
1	-1	1	4672	ultralytics.nn.modules.conv.Conv	[16, 32, 3, 2]
2	-1	1	7360	ultralytics.nn.modules.block.C2f	[32, 32, 1, True]

Testing/Result :



Analyze image

IMAGE UPLOAD

Browse

Drag and drop, browse, or paste your image.

PNG, JPG, WebP, and GIF files are supported. The preview appears instantly so you can verify the input before analysis.



Preview ready

Review the selected image, then run detection to generate the object list and analysis summary.

Remove image

WEBCAM CAPTURE

Start Camera

Use your camera for a live scan.

Start the camera, capture a frame, and send it into the same detection flow as uploaded images.

Camera is idle

Capture

Retake

If the backend returns an error, we surface it here so the user can retry immediately.

Total Waste Detected

13

Items present in the scan

Total Objects

13

Detected object count

Dominant Waste

cardboard

Most common class

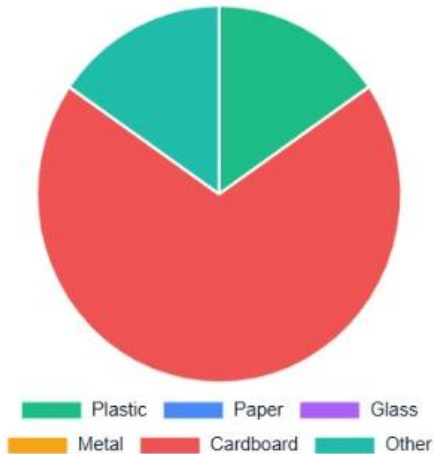
Sustainability Score

69%

More recyclable is better

WASTE BREAKDOWN**Closed pie chart by category**

A full pie chart keeps the complete composition visible without switching to a donut or gauge.

**TOTAL**
13

Plastic	2
Paper	0
Glass	0
Metal	0
Cardboard	9
Other	2

AI SUSTAINABILITY ASSISTANT

Get Groq-powered disposal, reduce, reuse, recycle, and recover suggestions from the dedicated Reuse & Recycle page.

[Open AI Guidance →](#)**DIY REUSE IDEAS**

Generate three Pollinations-based DIY ideas from the detected waste type when you want hands-on reuse inspiration.

[Open DIY Ideas →](#)**NEARBY RECYCLING CENTERS**

Use your location on the Centers page to find recycling centers, view them on the map, and open directions.

DIY REUSE IDEAS

Generate three projects only when you want them

This keeps the page responsive and avoids automatic image generation after detection.

▶ Show DIY Reuse Ideas



GREEN REUSE IDEA

Storage Jar

DIY PROJECT

Storage Jar

Clean and reuse the item as a small home storage solution or desk accessory.



GREEN REUSE IDEA

Planter

DIY PROJECT

Planter

Give the object a second life as a decorative planter or garden helper.



GREEN REUSE IDEA

Organizer

DIY PROJECT

Organizer

Turn the waste item into a compact organizer for practical everyday use.

Find Recycling Centers ↻

↻ Back to analysis

LOCATION

Location granted

The browser permission prompt controls this step.

TARGET WASTE TYPE

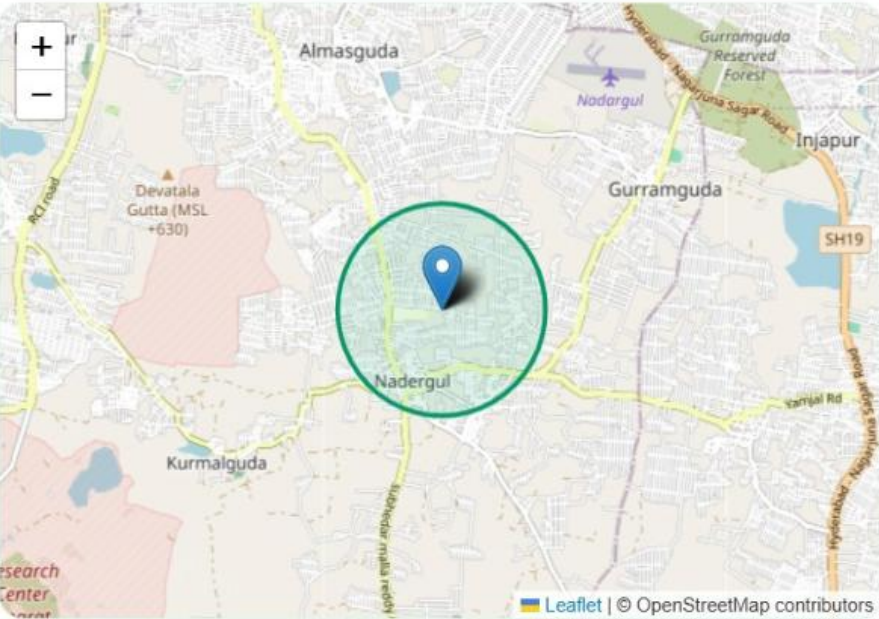
cardboard

Used to refine the search keywords sent to the backend.

CENTERS FOUND

4

Sorted by distance and shown with map links.



Paper Recycling Center

📍 Open a map search near your location

DISTANCE
Nearby

cardboard

🗑️ Maps

Paper Collection Center

📍 Open a map search near your location

DISTANCE
Search

cardboard

🗑️ Maps

cardboard

13 objects detected - 16/06/2026, 21:44:19

9 recyclable

Kandukur mandal
Leaflet | © OpenStreetMap contributors

WASTE REPORTS

Municipality queue

Most common waste: cardboard



trash

Reported by abc - 16/06/2026, 21:31:51

Location: 17.28187, 78.54279

In Progress

Open Location

Mark In Progress

Mark Resolved



cardboard

Reported by kavya - 16/06/2026, 19:34:49

Location: 17.28187, 78.54300

In Progress

Open Location

Mark In Progress

Mark Resolved

Back to analysis

View history

Future Scope :

- Integrate IoT-enabled smart bins for real-time waste monitoring and management.
- Develop a mobile application for easy access and on-the-go waste reporting.
- Enhance AI models to improve waste classification accuracy and support more waste categories.
- Collaborate with recycling centers and municipal authorities for automated waste collection and processing.

Conclusion :

The Smart Waste Analyzer provides an intelligent and user-friendly solution for effective waste management by combining AI-based waste classification, recycling guidance, and community participation. The system promotes proper waste segregation, supports sustainable practices, and assists municipal authorities in managing waste efficiently. By encouraging responsible disposal and recycling habits, the project contributes to a cleaner environment and a more sustainable future.